Euclid's Elements — Book II
Proposition 7

Formal Reconstruction — Technical Documentation

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CGJteamLab

If a straight line is cut at an arbitrary point, the square on the whole together with the square on either part equals twice the rectangle contained by the whole and that part, together with the square on the remaining part.

1 Introduction

Proposition II.7 returns from the midpoint identities II.5–II.6 to an arbitrary division point. Let C lie between A and B. Euclid's statement permits either of the two parts to be chosen as the "said segment." The current Lean theorem chooses AC. In geometric-content notation its conclusion is

$$\mathrm{Sq}(AB) + \mathrm{Sq}(AC) = 2\ \mathrm{Rect}(AB, AC) + \mathrm{Sq}(CB).$$

The modern algebraic shadow is immediate if $AC = a$ and $CB = b$:

$$(a + b)^2 + a^2 = 2(a + b)a + b^2.$$

This formula is only a mnemonic for the geometric pattern. Neither Euclid's proposition nor the current Lean theorem defines a numerical multiplication of segment lengths. The formal objects are concrete square and rectangle representatives, and the conclusion is an exact `HilbertScissorsEq`.

The production source begins with

```
import CGJteamLab.Proposition2_4
```

so Proposition II.4 is the immediate module dependency. At theorem-call level the proof uses `euclid_proposition_2_4` and `euclid_proposition_2_3`. It does not call II.1 or II.2 directly; their rectangle-dissection machinery is inherited through II.4.

2 Euclid's Statement and Classical Configuration

2.1 The statement

Euclid considers a straight line cut at an arbitrary point. The square on the whole together with the square on one chosen part equals twice the rectangle contained by the whole and that chosen part, together with the square on the remaining part [?].

A common classical presentation chooses BC as the selected part and writes

$$\mathrm{Sq}(AB) + \mathrm{Sq}(BC) = 2\ \mathrm{Rect}(AB, BC) + \mathrm{Sq}(CA).$$

The current Lean theorem instead chooses AC:

$$\mathrm{Sq}(AB) + \mathrm{Sq}(AC) = 2\ \mathrm{Rect}(AB, AC) + \mathrm{Sq}(CB).$$

These are the two symmetric readings already allowed by the wording of the proposition. The formal theorem therefore does not alter the mathematical content; it fixes one of Euclid's two admissible choices.

2.2 The classical diagram

In the classical proof one constructs the square on AB, draws the diagonal, and then draws parallels through the division point C. The resulting configuration contains two complementary parallelograms about the diagonal, a gnomon, and the squares on the two parts. Euclid uses I.43 to compare the complements and then adds the same figures to equal figures until the desired whole-square identity is obtained [?].

The current formal proof does not reproduce that gnomon bookkeeping line-by-line. The first figure therefore records the classical geometric state itself: the square on AB, its diagonal, and the two parallels through the division point and the diagonal intersection. The two normal forms actually consumed by Lean are shown later, at the proof steps where they enter.

Diagrammatic audit. The classical construction deserves its own figure because the diagonal and parallel lines are genuine geometric data in Euclid's proof. The formal proof then changes viewpoint: its geometrically meaningful states are the II.4 square dissection and the cyclic reorientation needed before the II.3 call. Betweenness reversal, endpoint-orientation changes of an unchanged rectangle, and the final associative-commutative regrouping do not create new geometry and are therefore not given separate figures.

3 Classical Proof

A convenient reading of Euclid's proof takes the selected part to be BC. Construct the square on AB and the usual parallels through C. The diagonal of the square produces complementary parallelograms; I.43 identifies the relevant pair. After adding the same adjacent rectangle to both, Euclid obtains two equal larger figures, so their sum is twice one of them.

That doubled figure is identified with twice the rectangle contained by AB and BC, because one adjacent side in the constructed rectangle is equal to BC. The remainder of the large square is organized as a gnomon together with the square on BC. Finally, adding the square on the

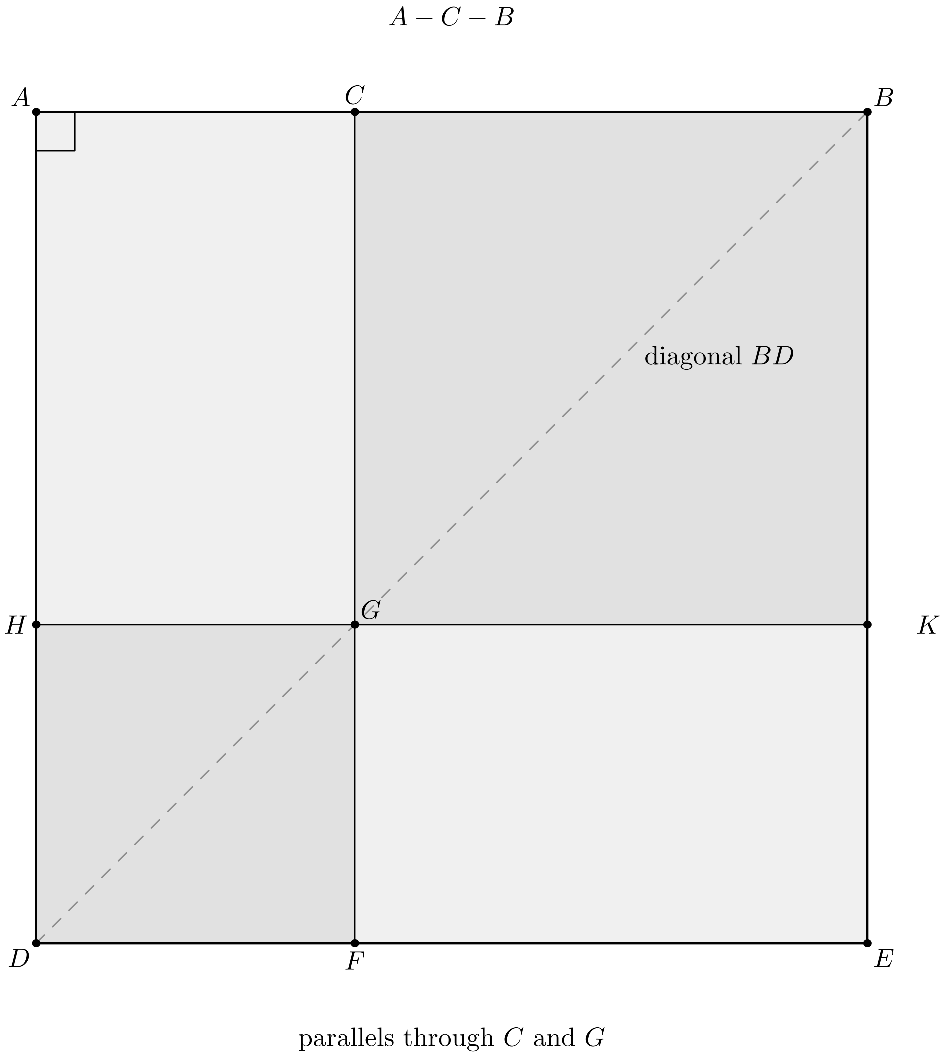

Figure 1: The classical square construction underlying Euclid II.7. The arbitrary cut is $A - C - B$; the diagonal BD and the parallels through C and G create the complementary parallelogram configuration used in the gnomon argument. The light shading separates the four regions without assigning them numerical areas. This is the source geometry; the current Lean proof later consumes already proved II.4 and II.3 normal forms instead of replaying this gnomon bookkeeping.

remaining part AC converts that gnomon expression into the whole square on AB plus the square on BC. Thus

$$\mathrm{Sq}(AB) + \mathrm{Sq}(BC) = 2\ \mathrm{Rect}(AB, BC) + \mathrm{Sq}(AC).$$

The classical local dependency picture is therefore schematically

$$\begin{gathered} \text{I.46 and I.31: construct square and parallels} \\ \Downarrow \\ \text{I.43: complementary parallelograms are equal} \\ \Downarrow \\ \text{gnomon and common-figure additions} \\ \Downarrow \\ \text{II.7.} \end{gathered}$$

The present Lean proof has a different DAG. It consumes already formalized Book II identities instead of reconstructing the gnomon from Book I each time.

4 The Geometric Identity in the Current Formalization

The baseline assumption is

```
(A B C : Geo.Point)
(hACB : Geo.Between A C B)
```

so the formal order is exactly

$$A - C - B.$$

No midpoint condition is present. This is the main geometric difference from II.5 and II.6.

The public conclusion is

$$\begin{aligned} &\mathrm{Sq}(AB) + \mathrm{Sq}(AC) \\ &\quad \simeq_{\mathrm{sc}} (\mathrm{Rect}(AB, AC) + \mathrm{Rect}(AB, AC)) + \mathrm{Sq}(CB). \end{aligned}$$

The notation 2 $\mathrm{Rect}(AB, AC)$ in the prose means two copies in a formal scissors sum. There is no scalar multiplication operation on content terms in the proof.

The actual Lean conclusion is even more concrete. The square on AB is the supplied square ABE_0D_0; the square on AC is supplied in reversed orientation as $CAFG$; the square on CB is $CBHK$; and the two final rectangles on AB, AC are literally the same supplied representative $U_0U_1U_2U_3$ written twice.

5 Reusable Book II Infrastructure Used

5.1 Proposition II.4: expand the square on the whole

The first normal form comes directly from `euclid_proposition_2_4`. With

$$W := \mathrm{Sq}(AB), \qquad X := \mathrm{Rect}(AC, CB), \qquad S := \mathrm{Sq}(AC), \qquad T := \mathrm{Sq}(CB),$$

the proof obtains

$$\boxed{W \simeq_{\mathrm{sc}} (X + S) + (X + T).}$$

A deliberate implementation choice makes this especially useful: the same concrete representative $T_0T_1T_2T_3$ of $\mathrm{Rect}(AC, CB)$ is supplied to both cross-rectangle positions inside II.4. Thus the two occurrences of X are literally the same scissors term, not merely representatives later proved equivalent.

5.2 Proposition II.3: normalize the rectangle on the whole and one part

The second normal form is

$$\boxed{R \simeq_{\mathrm{sc}} X + S,}$$

where

$$R := \mathrm{Rect}(AB, AC).$$

It comes from `euclid_proposition_2_3`, but not in the original orientation $A - C - B$. The proof first reverses the baseline to

$$B - C - A$$

and applies II.3 to that divided segment.

This reversal is geometric bookkeeping, not algebra. The contained-by data for R are reoriented from (AB, AC) to (BA, CA) by congruence symmetry. The cross rectangle is then rotated by `rectangle_contained_by_swap`, producing both the contained-by data needed by II.3 and an exact scissors equality between the rotated and original representatives.

5.3 The rectangle swap is geometric, not multiplicative commutativity

The call

```
have hSwapResult :=
  rectangle_contained_by_swap
    Geo
    T0 T1 T2 T3
    A C C B
    hCross
```

is the key representation step. It does not say that a numerical product $AC \cdot CB$ is commutative. It says that cyclically rotating a concrete rectangle changes the ordered side presentation while preserving its exact scissors term.

This is precisely the distinction established in the Book II rectangle API: geometric side-order transport precedes, and is independent of, any later arithmetic of segment lengths.

6 Proof Architecture of the Reconstruction

Unlike II.6, the production file introduces no private proposition-local helper declarations. The whole reconstruction is one public theorem containing local `let` abbreviations and `have` steps. The proof architecture is therefore visible directly in the theorem body.

6.1 Step 1: name the five scissors terms

The proof introduces

$$\begin{array}{rcl} W & := & \mathrm{Sq}(AB), \\ R & := & \mathrm{Rect}(AB, AC), \\ X & := & \mathrm{Rect}(AC, CB), \\ S & := & \mathrm{Sq}(AC), \\ T & := & \mathrm{Sq}(CB). \end{array}$$

These are abbreviations for supplied `HilbertScissorsTerm` objects, not new geometric constructions.

6.2 Step 2: obtain the II.4 normal form

The local fact `hW` calls II.4 and proves

$$W \simeq_{\mathrm{sc}} (X + S) + (X + T).$$

The same X representative is passed twice to the II.4 call.

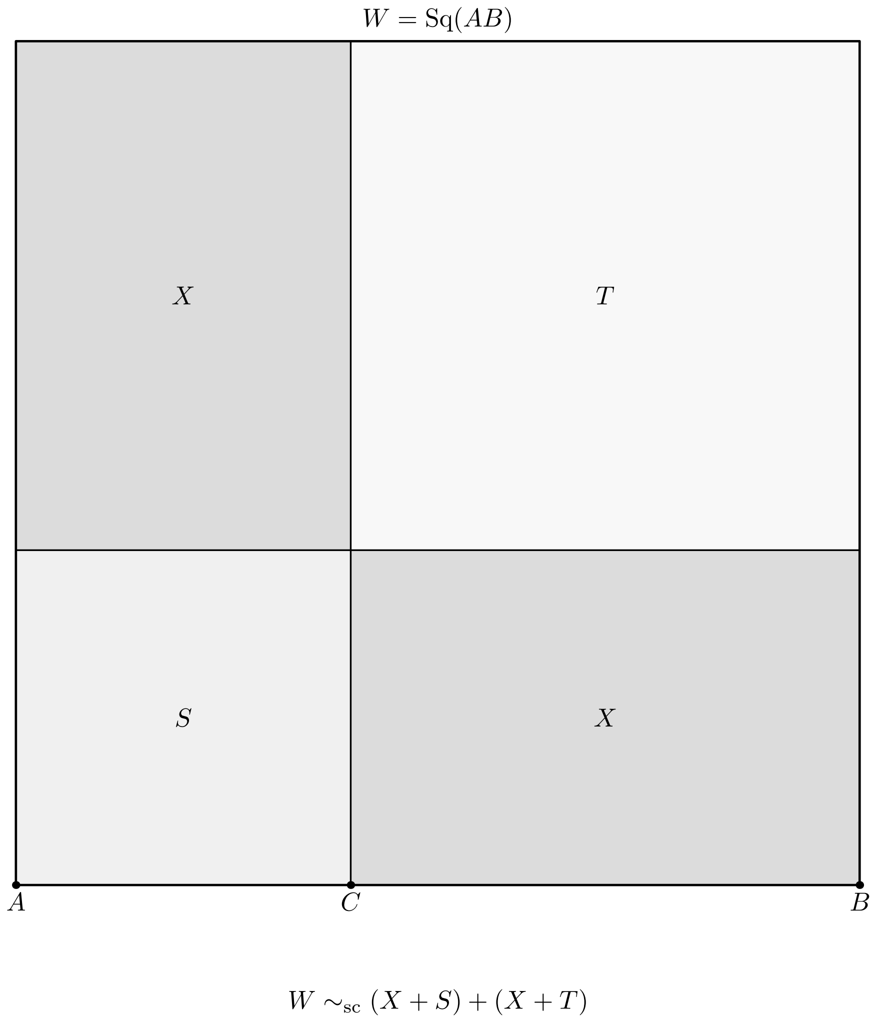

Figure 2: The first formal geometric state used by II.7. Proposition II.4 decomposes the supplied square W on AB at the cut C into the square term S on AC, the square term T on CB, and two literal copies of the same cross-rectangle term X. Thus the picture encodes $W \simeq_{\mathrm{sc}} (X + S) + (X + T)$ as an actual square dissection. No final II.7 regrouping is shown here.

6.3 Step 3: reverse the betweenness orientation

From

$$A - C - B$$

the proof derives

$$B - C - A$$

using `HilbertOrder.between_incidence`. This is the exact order required by the chosen II.3 instantiation.

6.4 Step 4: reorient the contained-by data for R

The representative $U_0U_1U_2U_3$ initially satisfies

$$\mathrm{Rect}(AB, AC).$$

For the II.3 call its side data are re-read as

$$\mathrm{Rect}(BA, CA).$$

The underlying rectangle is unchanged; only the orientations of the two unoriented segment congruences are reversed with `CongruentSwapSecond`.

6.5 Step 5: rotate the cross rectangle

The call to `rectangle_contained_by_swap` produces a cyclic rotation

$$T_0T_1T_2T_3 \longmapsto T_1T_2T_3T_0.$$

The rotated rectangle can then be presented with the ordered containing sides required by II.3. The same call also supplies an exact scissors equality between the two cyclic presentations.

6.6 Step 6: apply II.3 on $B - C - A$

The local fact `hRawR` proves

$$R \simeq_{\mathrm{sc}} X_{\mathrm{rot}} + S.$$

The subsequent facts `hRotateCrossBack` and `hR_concrete` transport this to the original cross representative:

$$\boxed{R \simeq_{\mathrm{sc}} X + S.}$$

No representative-uniqueness theorem is needed. The proof knows exactly how the rotated rectangle is related to the original one.

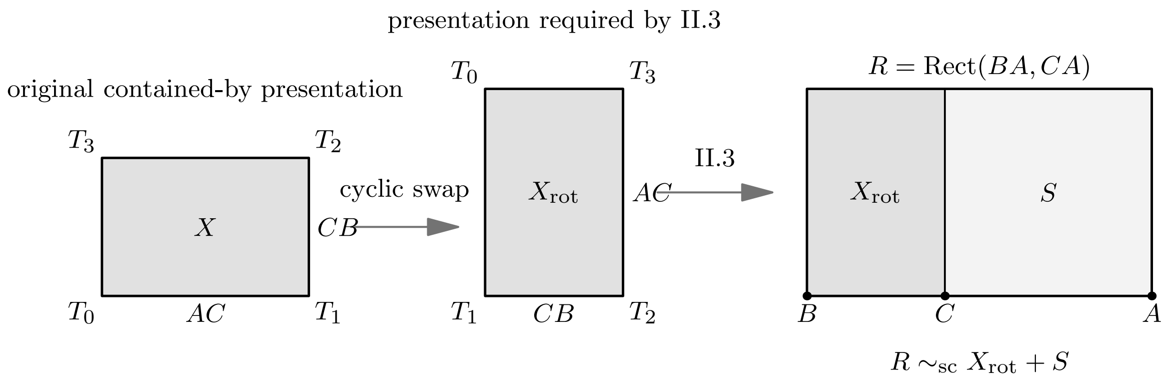

Figure 3: The second formal geometric state. On the left, the concrete cross rectangle X is shown with its original ordered side presentation. The cyclic swap changes only the vertex presentation and exchanges the ordered containing sides, producing the form required by II.3. On the right, II.3 is applied to the reversed baseline $B - C - A$: the rectangle R contained by BA, CA is split at C into the rotated cross rectangle and the square term S on CA. This is the geometric content behind $R \simeq_{\mathrm{sc}} X + S$ after transport back to the original cross representative.

6.7 Step 7: add the extra square S

From the II.4 normal form, `hAddS` gives

$$W + S \simeq_{\mathrm{sc}} ((X + S) + (X + T)) + S.$$

6.8 Step 8: regroup the formal sum

The only literal equality of terms used in the final bridge is

$$((X + S) + (X + T)) + S = ((X + S) + (X + S)) + T.$$

Lean proves it by

```
have hRegroup :
    ((X + S) + (X + T)) + S
      =
    ((X + S) + (X + S)) + T := by
  ac_rfl
```

so this step uses only associativity and commutativity of the formal sum.

6.9 Step 9: replace the two copies of $X + S$

Since `hR` states

$$R \simeq_{\mathrm{sc}} X + S,$$

its symmetry can replace each copy of $X + S$ by R. The proof uses `HilbertScissorsEq.add` twice and obtains

$$((X + S) + (X + S)) + T \simeq_{\mathrm{sc}} (R + R) + T.$$

6.10 Step 10: compose

Transitivity finally gives

$$\boxed{W + S \simeq_{\mathrm{sc}} (R + R) + T.}$$

Expanding the abbreviations is exactly the public II.7 conclusion.

This proof is additive from beginning to end. It never cancels a common content term, subtracts one figure from another, or invokes a positivity principle.

Diagrammatic audit. No additional picture is introduced for Steps 7–10. Once the II.4 state in Figure 2 and the II.3 state in Figure 3 are available, the remaining proof consists of adding the already supplied square term, reassociating and commuting a formal sum, replacing two exact-scissors-equal subterms, and composing the resulting relations. Drawing a new region arrangement for those operations would suggest a new physical dissection that the Lean proof does not construct.

7 Lean Formalization

7.1 The public theorem signature

The following is the current compiling public signature copied from `CGJteamLab/Proposition 2_7.lean`. The amount of witness data reflects the fact that II.7 reuses the configured II.4 and II.3 interfaces rather than hiding their geometry behind an existential wrapper.

```
theorem euclid_proposition_2_7
    [HilbertIncidence Geo]
    [HilbertEuclideanPlane Geo]

    (A B C : Geo.Point)

    ------------------------------------------------------------------
    -- II.2 / II.4 diagram: square on AB cut at C.
    ------------------------------------------------------------------

    (D0 E0 L0 X0 : Geo.Point)
    (hSquareAB : IsSquare Geo A B E0 D0)

    (hACB : Geo.Between A C B)
    (hD0L0E0 : Geo.Between D0 L0 E0)
    (hD0X0B : Geo.Between D0 X0 B)
    (hCX0L0 : Geo.Between C X0 L0)

    (hII2LeftPar :
      IsParallelogram Geo L0 D0 A C)
    (hII2RightPar :
      IsParallelogram Geo B E0 L0 C)

    -- Representatives of Rect(AB,AC) and Rect(AB,CB) used by II.4.
    (U0 U1 U2 U3 : Geo.Point)
    (V0 V1 V2 V3 : Geo.Point)

    (hAB_AC :
      IsRectangleContainedBy Geo
        U0 U1 U2 U3 A B A C)

    (hAB_CB :
      IsRectangleContainedBy Geo
        V0 V1 V2 V3 A B C B)

    ------------------------------------------------------------------
    -- II.3 diagram for the left II.4 term, read on B-C-A.
    ------------------------------------------------------------------

    (D1 E1 L1 X1 : Geo.Point)

    (hRectLeft :
      IsRectangle Geo B A E1 D1)

    (hAE1_CA :
      Geo.Congruent A E1 C A)

    (hD1L1E1 : Geo.Between D1 L1 E1)
```

```
  (hD1X1A : Geo.Between D1 X1 A)
  (hCX1L1 : Geo.Between C X1 L1)

  (hLeftLeftPar :
    IsParallelogram Geo L1 D1 B C)
  (hLeftRightPar :
    IsParallelogram Geo A E1 L1 C)

  -- Square on CA, i.e. the same unoriented segment as AC.
  (F G : Geo.Point)
  (hSquareCA : IsSquare Geo C A F G)

  ----------------------------------------------------------------
  -- II.3 diagram for the right II.4 term.
  ----------------------------------------------------------------

  (D2 E2 L2 X2 : Geo.Point)

  (hRectRight :
    IsRectangle Geo A B E2 D2)

  (hBE2_CB :
    Geo.Congruent B E2 C B)

  (hD2L2E2 : Geo.Between D2 L2 E2)
  (hD2X2B : Geo.Between D2 X2 B)
  (hCX2L2 : Geo.Between C X2 L2)

  (hRightLeftPar :
    IsParallelogram Geo L2 D2 A C)
  (hRightRightPar :
    IsParallelogram Geo B E2 L2 C)

  -- Square on CB.
  (H K : Geo.Point)
  (hSquareCB : IsSquare Geo C B H K)

  ----------------------------------------------------------------
  -- One representative of Rect(AC,CB), used twice in II.4.
  ----------------------------------------------------------------

  (T0 T1 T2 T3 : Geo.Point)

  (hCross :
    IsRectangleContainedBy Geo
      T0 T1 T2 T3 A C C B) :

  HilbertScissorsEq Geo
    (hilbertParallelogramTerm Geo A B E0 D0 +
     hilbertParallelogramTerm Geo C A F G)
    ((hilbertParallelogramTerm Geo U0 U1 U2 U3 +
      hilbertParallelogramTerm Geo U0 U1 U2 U3) +
     hilbertParallelogramTerm Geo C B H K)
```

The signature has four geometric blocks.

1. The points A, B, C and `hACB` specify the arbitrary internal cut.

2. The D_0, E_0, L_0, X_0 block supplies the square-on-AB diagram required by II.4, together with representatives of $\mathrm{Rect}(AB, AC)$ and $\mathrm{Rect}(AB, CB)$.
3. The D_1, E_1, L_1, X_1 and D_2, E_2, L_2, X_2 blocks are the two II.3 subconfigurations consumed inside II.4, together with the squares on CA and CB.
4. The T_0, T_1, T_2, T_3 block supplies one representative of $\mathrm{Rect}(AC, CB)$, deliberately reused twice.

The conclusion contains only the square on AB, the square on AC, two copies of the same rectangle on AB, AC, and the square on CB. Much of the signature is therefore construction witness data inherited from the reusable Book II APIs rather than conceptual data in the final identity.

7.2 The two decisive local facts

The mathematical core of the production proof is compact:

```
have hW :
    HilbertScissorsEq Geo
      W
      ((X + S) + (X + T)) := by
  ...
  exact euclid_proposition_2_4 ...

...

have hR :
    HilbertScissorsEq Geo
      R
      (X + S) := by
  ...
```

Everything after these two normal forms is an exact-scissors rearrangement. This is the central architectural fact of II.7.

8 Relationship with Earlier Book II Propositions

8.1 Proposition II.3

II.3 is called directly on the reversed divided segment $B - C - A$. Its role is to convert the rectangle contained by the whole AB and the selected part AC into

$$\mathrm{Rect}(AC, CB) + \mathrm{Sq}(AC).$$

Because the current II.3 interface has ordered contained-by side data, the proof must explicitly reverse segment orientations and rotate the cross rectangle before the call.

8.2 Proposition II.4

II.4 is the main normal-form theorem:

$$\mathrm{Sq}(AB) \simeq_{\mathrm{sc}} (\mathrm{Rect}(AC, CB) + \mathrm{Sq}(AC)) + (\mathrm{Rect}(AC, CB) + \mathrm{Sq}(CB)).$$

This is exactly the decomposition needed for II.7 once one recognizes the first parenthesis as $\mathrm{Rect}(AB, AC)$ by II.3.

8.3 Propositions II.1 and II.2

Neither `euclid_proposition_2_1_two` nor `euclid_proposition_2_2` is called directly by II.7. They are, however, below II.4 in the established Book II dependency chain. Thus II.7 inherits their rectangle split and square decomposition indirectly.

This distinction matters: the module/theorem DAG should not be replaced by a vague statement that II.7 "uses all previous propositions."

8.4 Propositions II.5 and II.6

II.5 and II.6 are not dependencies of II.7. They form the preceding midpoint block, whereas II.7 again assumes only an arbitrary cut. The production module therefore imports II.4 directly rather than continuing the linear module chain through II.6.

9 Relationship with Book I Infrastructure

Classically, II.7 is a gnomon argument centered on I.43 and on the square and parallel constructions supplied by I.46 and I.31. The current Lean theorem does not call those Book I propositions locally. Their geometric consequences have already been absorbed into the square, rectangle, parallelogram, and scissors infrastructure used by II.3 and II.4.

This is another clear example of the difference between the classical DAG and the formal library DAG. The reconstruction proves the same content identity without requiring every proposition to replay Euclid's local construction.

10 Formalization Notes and Hidden Geometric Data

10.1 The chosen segment differs from a common printed diagram

Many printed versions instantiate the proposition with BC as the selected part. The current Lean theorem selects AC. Because Euclid's statement says "one of the segments," this is not a mathematical strengthening or weakening. It is a concrete orientation choice that must nevertheless be recorded in technical documentation.

10.2 Betweenness reversal is explicit

The source assumption is `Geo.Between A C B`. II.3 is used on $B - C - A$, so the reversed betweenness statement is derived explicitly. The formal proof cannot infer the desired orientation merely from the picture.

10.3 Squares use unoriented side congruence

The final square on the selected part is supplied as

```
(hSquareCA : IsSquare Geo C A F G)
```

although the prose calls it the square on AC. This is legitimate because the underlying segment is unoriented at the congruence level, but the orientation is visible in the concrete Lean arguments.

10.4 The unused final representative V

The II.4 interface requires representatives of both $\text{Rect}(AB, AC)$ and $\text{Rect}(AB, CB)$. The latter, represented by $V_0V_1V_2V_3$, is internal to the II.4 call and does not appear in the final II.7 conclusion. This is an example of witness data required by a reused theorem but not by the mathematical endpoint.

10.5 Representative reuse avoids uniqueness transport

The same T rectangle is supplied twice to II.4. The same U rectangle is written twice in the final target. These choices make repeated terms literal. The proof therefore does not call `rectangle_contained_by_unique`.

This is not a restriction on the proposition. It is a proof-engineering normalization that removes unnecessary representative bookkeeping.

10.6 No cancellation principle

The algebraic shadow strongly invites subtraction. The Lean proof does not subtract X, S, or any square. It adds S, rearranges a commutative formal sum, and replaces equal-scissors subterms. No proper-part principle, positivity theorem, or content cancellation axiom enters II.7.

11 Axiomatic and Semantic Status

The public theorem assumes

```
[HilbertIncidence Geo]
[HilbertEuclideanPlane Geo]
```

and the production file introduces no new proposition-specific `axiom`, `sorry`, or `admit`.

The user-verified transitive axiom audit is

```
#print axioms Geometry.euclid_proposition_2_7
```

```
'Geometry.euclid_proposition_2_7' depends on axioms: [propext,
 Classical.choice,
 Geometry.bookZero_nullSegment1,
 Quot.sound]
```

Hence the precise status is not "axiom-free." Proposition II.7 adds no new axiom and no `sorry`; its reported transitive dependencies are the four declarations displayed above, with no `sorryAx`.

The public module and the full relevant project build were reported clean after the theorem was added to the project. Semantically, the theorem lies in the exact finite-scissors layer. It proves `HilbertScissorsEq` and does not invoke a numerical area function, segment multiplication, or general equicomplementability as its conclusion.

12 Dependency Summary

The theorem-call DAG is

$$\begin{array}{c}
\text{Hilbert incidence + Euclidean plane layer} \\
\Downarrow \\
\text{Book I square, rectangle, parallelogram, and scissors infrastructure} \\
\Downarrow \\
\text{HilbertRectangle contained-by and swap API} \\
\Downarrow \\
\text{II.3 rectangle-plus-square} \quad + \quad \text{II.4 square expansion} \\
\Downarrow \\
\text{betweenness/orientation transport + rectangle rotation} \\
\Downarrow \\
\text{exact-scissors addition + AC regrouping} \\
\Downarrow \\
\text{public theorem } \texttt{euclid_proposition_2_7}.
\end{array}$$

The immediate module chain is simply

$$\text{Proposition2\_4 module} \longrightarrow \text{Proposition2\_7 module}.$$

The classical local chain is different:

$$\text{I.46} + \text{I.31} + \text{I.43} \longrightarrow \text{gnomon comparison} \longrightarrow \text{common-figure additions} \longrightarrow \text{II.7}.$$

New geometric axiom in II.7:	no
New content axiom in II.7:	no
New proposition-local helper declaration:	no; local `haves` only
New reusable public helper theorem:	no
New `HilbertRectangle` theorem:	no
New Interface / Book Zero / Scissors theorem:	no
Directly reuses earlier Book II theorem:	yes, II.3 and II.4
Directly calls II.1 or II.2:	no
Imports II.5 or II.6:	no
Uses exact `HilbertScissorsEq`:	yes
Uses general equicomplementability as conclusion:	no
Uses numerical area:	no
Uses segment multiplication:	no
Uses content cancellation/subtraction:	no
Uses representative uniqueness:	no
Public theorem compiles:	yes
Module build:	clean (user verified)
Full relevant project build:	clean (user verified)
Proposition-local `sorry`/`admit`:	no
Transitive `sorryAx`:	absent in reported audit

13 Conclusion

Proposition II.7 is an especially transparent demonstration of the reusable Book II rectangle calculus. The classical proof organizes the square on the whole through a gnomon and complementary parallelograms. The current Lean proof instead starts from two already established exact-scissors normal forms.

II.4 gives

$$W \simeq_{\mathrm{sc}} (X + S) + (X + T),$$

and II.3, after reversing the divided segment and rotating a rectangle representative, gives

$$R \simeq_{\mathrm{sc}} X + S.$$

Adding S, regrouping the formal sum by associativity and commutativity, and replacing two copies of $X + S$ by R yields

$$W + S \simeq_{\mathrm{sc}} (R + R) + T.$$

Thus the formal proof preserves the synthetic geometric-content character of Book II while avoiding numerical area, segment multiplication, subtraction, and cancellation. II.7 introduces no new axiom and no new public infrastructure; its value is architectural: it shows that II.3 and II.4 are already strong enough to recover another classical gnomon identity by exact scissors composition.